

Diesel Control Unit Local/Remote Switch Does Not Switch to Local

Symptoms

- Engine will **not** crank when the start button is pressed at the remote panel.
- Diesel control unit local/remote switch does **not** to switch to remote
 - Diesel control unit indicates incorrect assignment
 - Remote panel does **not** switch to local
 - Remote panel does **not** switch to remote
 - Remote panel indicates incorrect assignment.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

- To initiate engine crank from the remote panel, the following panel parameters **must** be met:
- The remote panel power lamp illuminated
 - The local start **only** lamp is **not** illuminated
 - The engine **must** be stopped.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box.	
	STEP 1A. Check the DCU410 unit display for faults.	
	STEP 1A-1. Check the DCU410 unit power supply wire for voltage +24-VDC.	

STEPS		SPECIFICATIONS
	STEP 1B. Check the function of the start button.	
	STEP 1C. Check the engine stop indication wire for an open.	
	STEP 1D. Check the ignition engine stop supply and engine stop switch wires for an open.	
	STEP 1E. Check the vessel remote data link Ethernet signal wire for an open.	
	STEP 1F. Check the engine stop indication wire for a wire-to-wire short.	
	STEP 1G. Check the ignition engine stop supply and engine stop switch wires for a wire-to-wire short.	
	STEP 1H. Check the vessel remote data link Ethernet signal wire for a wire-to-wire short.	
	STEP 1I. Check the engine stop indication wire for a short to ground.	
	STEP 1J. Check the ignition engine stop supply and engine stop switch wires for a short to ground.	
	STEP 1K. Check the vessel remote data link Ethernet signal wire for a short to ground.	

STEP 1. Check the customer interface box.

STEP 1A. Check the DCU410 unit display for faults.

Conditions :

- Locate the DCU410 unit display.

Action	Specification/Repair	Next Step
Check the DCU410 unit display for indication of faults.	DCU410 unit indicates fault(s)? YES	1B
	DCU410 unit indicates fault(s)? NO	1A-1

STEP 1A-1. Check the DCU410 unit power supply wire for voltage +24-VDC.**Conditions :**

- Open the customer interface box.

Action	Specification/Repair	Next Step
Check the voltage at the battery 1 voltage (switched power) at the DCU410 unit. • Place one test on the battery 1 voltage supply wire at the DCU410 unit. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC? YES Repair : Check the batteries. Refer to OEM service manual.	Repair complete
	Less than +24-VDC? NO	1B

STEP 1B. Check the function of the start button.**Conditions :**

- Open the customer interface box.
- Disconnect the local mode supply wire at the DCU410 unit, CLU, and X4 connections.

Action	Specification/Repair	Next Step
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Check the function of the stop button. - Place one test lead on the local mode supply wire at the DCU410 unit. - Place the other test lead on the local mode supply wire at the CLU unit. - Place one test lead on the local mode supply wire at the DCU410 unit. - Place the other test lead on the local mode supply wire at the X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1C
	Less than 10 ohms? NO Repair : Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete

STEP 1C. Check the engine stop indication wire for an open.

Conditions :

- Open the customer interface box.
- Disconnect the engine stop indication and energize to stop relay wires from the DCU410 unit and CLU unit.

Action	Specification/Repair	Next Step
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Check the engine stop indication wire for an open. - Place one test lead on the engine stop indication supply wire at the DCU410 unit. - Place the other test lead on the engine stop indication supply wire at the CLU unit. - Place one test lead on the energize to stop relay return wire at the DCU410 unit. - Place the other test on the energize to stop relay return wire at the CLU unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1D
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 1D. Check the ignition engine stop supply and engine stop switch wires for an open.

Conditions : - Open the customer interface box. - Disconnect the ignition engine stop supply and engine stop switch wires from the C1 connector and switch.		
Action	Specification/Repair	Next Step

Check the ignition engine stop supply and engine stop switch wires for an open. - Place one test lead on the ignition stop supply wire at the C1 connector. - Place the other test lead on the ignition stop supply wire at the engine stop switch. - Place one test lead on the engine stop switch wire at the DCU410 unit. - Place the other test lead on the engine stop switch wire at the engine stop switch. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1E
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the engine stop switch. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 1E. Check the vessel remote data link Ethernet signal wire for an open.

Conditions : - Open the customer interface box. - Disconnect the vessel remote data link Ethernet signal wire at the DCU410 unit and Ethernet switch.		
Action	Specification/Repair	Next Step

Check the vessel remote data link Ethernet signal wire for an open. - Place one test lead on the vessel remote data link Ethernet signal wire at the DCU410 unit. - Place the other test lead on the vessel remote data link Ethernet signal wire at the Ethernet switch. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1F
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 1F. Check the engine stop indication wire for a wire-to-wire short.

Conditions : Open the customer interface box.		
Action	Specification/Repair	Next Step

Check the engine stop indication wire for a wire to wire short. - Place one test lead on the engine stop indication supply wire at the DCU410 unit. - Place the other test lead on all wires at the DCU410 unit. - Place one test lead on the energize to stop relay return wire at the DCU410 unit. - Place the other test on all other wires at the DCU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1G
	Less than 10 ohms? NO Repair : Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete

STEP 1G. Check the ignition engine stop supply and engine stop switch wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step
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Check the ignition engine stop supply and engine stop switch wires for a wire-to-wire short. - Place one test lead on the ignition stop supply wire at the C1 connector. - Place the other test lead on all other wires at the C1 connector. - Place one test lead on the engine stop switch wire at the DCU410 unit. - Place the other test lead on all wires at the DCU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1H
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete

STEP 1H. Check the vessel remote data link Ethernet signal wire for a wire-to-wire short.

Conditions : Open the customer interface box.		
Action	Specification/Repair	Next Step

Check the vessel remote data link Ethernet signal wire for a wire-to-wire short. - Place one test lead on the vessel remote data link Ethernet signal wire at the DCU410 unit. - Place the other test lead on all other wires at the DCU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1I
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 1I. Check the engine stop indication wire for a short to ground.

Conditions : Open the customer interface box.		
Action	Specification/Repair	Next Step
Check the engine stop indication wire for a short to ground. - Place one test lead on the engine stop indication supply wire at the DCU410 unit. - Place the other test lead on panel ground. - Place one test lead on the energize to stop relay return wire at the DCU410 unit. - Place the other test on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1J
	Less than 10 ohms? NO Repair : Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete

STEP 1J. Check the ignition engine stop supply and engine stop switch wires for a short to ground.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step
Check the ignition engine stop supply and engine stop switch wires for a short to ground. • Place one test lead on the ignition stop supply wire at the C1 connector. • Place the other test lead on panel ground. • Place one test lead on the engine stop switch wire at the DCU410 unit. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1K
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete

STEP 1K. Check the vessel remote data link Ethernet signal wire for a short to ground.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step

Check the vessel remote data link Ethernet signal wire for a short to ground. - Place one test lead on the vessel remote data link Ethernet signal wire at the DCU410 unit. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	Contact a Cummins® Authorized Repair Location.
	Less than 10 ohms? NO Repair : Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete